

When Awareness Isn't Enough: Exploring Physical Interventions for Nail-Biting

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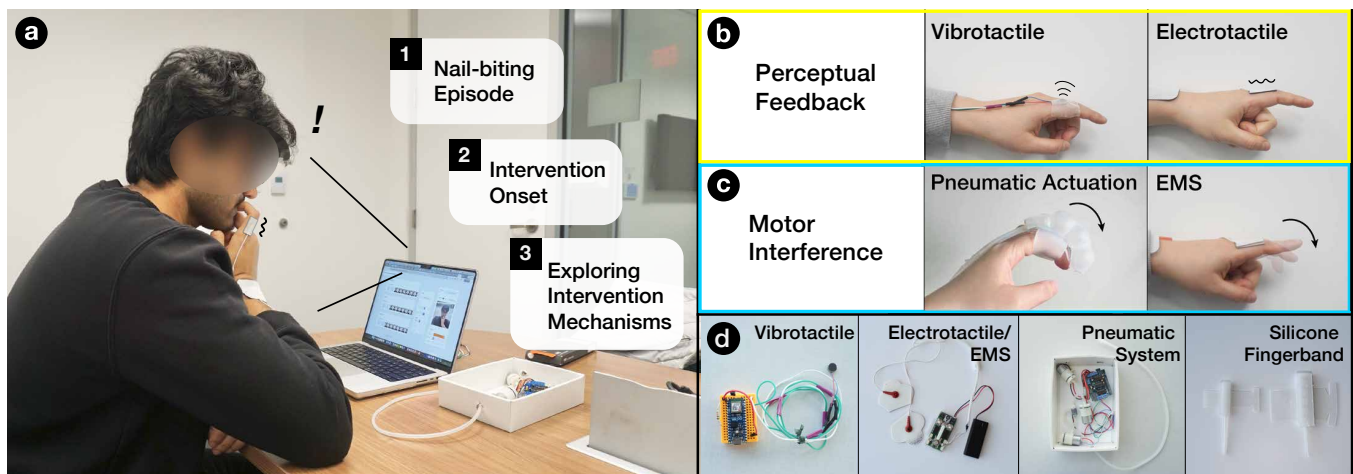


Figure 1: Overview of the study. (a) Study setup with participants simulating nail-biting. Two intervention mechanisms were explored: (b) perceptual feedback (vibrotactile, electrotactile) and (c) motor interference (pneumatic actuation, electrical muscle stimulation (EMS)). (d) These mechanisms were instantiated via four functional probes: vibrotactile, electrotactile, a pneumatic system with a custom-made silicone fingerband, and EMS.

Abstract

Nail-biting is a habitual behavior that often occurs automatically. Individuals with limited self-control struggle to stop the behavior even when they are aware of it, making existing awareness-based interventions insufficient. Recent work reports that such individuals desire more direct, physically interruptive interventions; however, how users experience and accept these interventions remains under-explored. We conducted exploration sessions with active nail-biters to examine two intervention mechanisms—perceptual feedback and motor interference—instantiated through four wearable probes: vibrotactile, electrotactile, pneumatic actuation, and electrical muscle stimulation (EMS). Interestingly, we observed that active nail-biters preferred motor interference and were willing to accept reduced

motor agency when they could retain autonomy over the intervention system. Our findings characterize how nail-biters negotiate agency in inhibiting nail-biting, highlighting the importance of user autonomy and offering insights for designing future nail-biting interventions.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Haptic devices*; *Ubiquitous and mobile devices*.

Keywords

Nail-biting, Body-focused Repetitive Behaviors, Onychophagia, Wearable Interventions, Haptic Feedback, Motor Interference, Sense of Agency

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1 Introduction

Onychophagia, a type of Body-Focused Repetitive Behaviors (BFRBs), is a repetitive behavior where 20–30% of the general population compulsively bite or chew their nails and nail beds [9]. It can cause physical harm such as inflamed fingers and dental problems [2, 12], and is often accompanied by social stigma, with nail-biters perceived as immature or nervous [31]. To inhibit nail-biting, clinical and HCI research has largely centered on awareness-based approaches. One of the most notable approaches is Habit Reversal Training (HRT) [1, 4], which focuses on making individuals aware of their behavior to enable them to stop voluntarily [21]. Commercial and research wearables support this voluntary inhibition by detecting nail-biting and alerting users upon detection through haptic notifications, such as vibrotactile or electrotactile feedback [11, 14, 17, 23].

However, as nail-biting is often an automatized habit, it can be initiated and executed without conscious awareness [19, 30]. Even when individuals are aware of the ongoing behavior, some struggle to inhibit due to limited self-control, making perceptual feedback alone insufficient [13, 28]. As an alternative to awareness-based interventions, researchers have also employed physical prevention methods. Passive barriers such as gloves and dental appliances block finger access to the mouth [2, 20, 27], making it impossible to bite. However, they require constant wear and interfere with everyday hand use. Similarly, bitter-tasting nail polish, an aversive method, has been shown to reduce nail-biting [15]. However, it poses adherence challenges, as it requires continuous wear, which users cannot control when the aversive feedback occurs. These limitations motivate nail-biting interventions that act only during nail-biting moments, ranging from perceptual feedback to assertive physical methods.

Interestingly, a recent study reports that nail-biters who have limited self-control actively desire more direct, physically interruptive interventions [25]. Yet, prior work has not explored this stated preference and how active nail-biters perceive and experience interventions with different underlying mechanisms, nor how considerations of impact, agency, and autonomy shape these user experiences. Understanding these experiences from active nail-biters' perspectives is essential for designing wearable interventions that are both effective and acceptable, helping inform responsible HCI design by clarifying boundaries of control and comfort that support long-term adherence.

To address this gap, we conceptualize nail-biting interventions along two mechanisms: *perceptual feedback* (Figure 1b), which provides perceptual cues to support awareness of nail-biting behavior, and *motor interference* (Figure 1c), which introduces resistive motor interference to make hand-to-mouth motion difficult. Drawing on existing haptic techniques commonly used in HCI, we instantiated these mechanisms with four wearable probes (Figure 1d). We then conducted one-on-one exploration sessions with active nail-biters to explore how these mechanisms are experienced in participants' lived experiences of nail-biting. We make three contributions: (i) characterization of how perceptual feedback and motor interference shape user experiences; (ii) observations suggesting that active nail-biters accepted reduced motor agency while emphasizing the importance of retaining autonomy over the intervention

system; (iii) early design implications that emerge from exploration with active nail-biters.

2 System Implementation

2.1 Probe Implementations

We built four wearable probes designed as functional instantiations of intervention mechanisms spanning perceptual feedback and motor interference (Figure 1b-d).

Perceptual Feedback. We selected **vibrotactile** feedback as a baseline perceptual cue because it is the most common and socially familiar modality in everyday wearables [3, 11]. In our implementation, vibrotactile feedback was delivered via a coin motor (10 mm×3 mm, 3V DC, ~12,000 rpm) controlled by an Arduino Nano. The motor was mounted on the finger and set to a familiar intensity comparable to smartphone notifications [10, 29]. We included **electrotactile** feedback to represent more salient perceptual feedback than vibration, often perceived as tingling sensations [33]. It has also been used in commercial habit-breaking devices [23]. Surface gel electrodes were driven by a custom stimulation device (pulse width: 0–500 μ s, frequency: 30 Hz, up to 35V) [22]. Intensity was individually calibrated below the pain threshold while ensuring clear perceptibility without involuntary movement.

Motor Interference. We used **pneumatic actuation** to introduce soft, compliant resistance that made finger movement more effortful during hand-to-mouth actions, while still allowing participants to complete the movement voluntarily [8, 24, 32]. A custom-fabricated soft silicone fingerband made with Ecoflex 00-50 was connected to a compact air-pump system (3× micro pumps, 4.5V, ~2.5 LPM) controlled via an Arduino Uno. We employed **electrical muscle stimulation (EMS)** to directly activate finger muscles and induce involuntary finger flexion, disrupting the hand posture required to complete the hand-to-mouth action [18, 22]. The same stimulation hardware as the electrotactile condition was used, with electrodes positioned near the proximal phalanx [16]. Intensity was gradually calibrated per participant to produce observable movement while maintaining safety and comfort.

2.2 Intervention Timing During Exploration

We used an image-based machine learning model from Teachable Machine [7] to ensure comparable intervention timing across participants. The model distinguished hand-to-mouth actions from other hand-to-face movements (e.g., resting the chin or touching hair). We placed a camera in front of participants and triggered interventions when the model detected simulated nail-biting behavior. This image-based setup enabled interaction with the wearable probes without attaching additional motion sensors, which could have disrupted the device experience.

3 Exploratory Study

We conducted one-on-one exploration sessions with active nail-biters (Figure 1a). To respect privacy around this stigmatized habit, each session paired one investigator with one participant in a private setting [5]. Participants then simulated their nail-biting behavior to experience all four probes. This study design enabled

controlled within-subject elicitation of experiences across multiple probes within a single session, which would be difficult to achieve with naturally occurring nail-biting episodes due to their sporadic and highly individualized nature in triggers and behavior patterns [30].

3.1 Participants

Nine participants were recruited via campus recruitment channels (5 female, 4 male; $M = 25.5$ years, $SD = 5.36$) and self-identified as having a severe nail-biting problem. Their nail-biting damage was visually confirmed, including significantly shortened nails, irregular nail edges, hangnails, and damaged cuticles. Three participants reported being able to exercise self-control to stop nail-biting upon becoming aware of their behavior, while the remaining 6 continued to bite despite awareness. This study was approved by the University Institutional Review Board (IRB 2076862-4). All participants provided written informed consent and received \$30 compensation. Sessions were audio- and video-recorded and transcribed.

3.2 Procedure

After a brief introduction, participants experienced the four probes in randomized order. In front of a laptop camera, participants simulated their hand-to-mouth nail-biting behavior while reflecting on their habitual contexts. All probes targeted the fingers and were individually calibrated to ensure safety. Electrotactile intensity was adjusted to produce salient sensations without pain or involuntary movement, while EMS was calibrated to elicit visible movement. Participants were provided with explicit opt-out controls and could immediately stop any intervention if they experienced discomfort or wished to discontinue.

After each probe, participants briefly reflected on their experience. Following the exploration of all four probes, we conducted a semi-structured interview. Participants rated each probe's perceived impact on nail-biting urges and reflected on their preferences between perceptual feedback versus motor interference as intervention mechanisms. They also discussed desired functions for future devices, usage contexts, and willingness for daily use. Two investigators used reflexive thematic analysis to independently code the data and collaboratively refine their interpretations [6], deriving design insights.

4 Findings

4.1 Perceived Impact on Nail-Biting Urges

Participants reflected on how each modality would affect their nail-biting urges in real-world situations based on simulated interactions. As shown in Figure 2, motor interference modalities (EMS: $M = 6.6$, $SD = 0.73$; Pneumatic: $M = 5.9$, $SD = 0.78$) were rated notably higher than perceptual feedback (Electrotactile: $M = 5.2$, $SD = 1.20$; Vibrotactile: $M = 3.6$, $SD = 1.59$).

Within motor interference, participants anticipated EMS would serve as a strong deterrent that they described as jolting but not painful, making hand-to-mouth actions challenging. Pneumatic actuation was valued for its soft barrier effect, restricting finger movement through what one participant described as "new" (P7) pressure sensations, although some felt that the mild pressure could be overcome. Within perceptual feedback, electrotactile provided

a "vibration-like" (P3) sensation that was more noticeable than vibrotactile. However, some participants found it unpleasant and associated it with punishment, with P5 noting, "*the next time I attempt to bite my nails, it will trigger memories of a negative experience.*" Vibrotactile feedback was described as familiar and easy to miss due to frequent exposure in everyday contexts such as "*phone and bus*" (P7). Several participants reported that it would be insufficient during strong urges because it did not interfere with movement.

4.2 Preferences for Motor Interference and Perceptual Feedback

Beyond probe-level differences, participants reflected on the underlying intervention mechanisms across the four probes. Most participants (8/9) expressed a preference for probes that introduced motor interference when reflecting on their experiences across the four modalities. Participants reported that motor interference would actively discourage continuation of the nail-biting, with P1 noting "*not worth fighting for.*" In contrast, participants proactively expressed the necessity of this level of intervention, as P3 noted, "*trying to stop nail-biting is stressful.*" Participants also noted limitations of existing methods, such as gel nail polish. While gel nails function as a physical barrier, they are costly, time-consuming to apply, and damaging to the nails. Most importantly, they do not interrupt the hand-to-mouth action itself. This limitation reinforced participants' interest in wearable interventions that directly interfere with the action.

4.3 Active Nail-Biters' Reported Considerations for Everyday Use

Acceptability of Reduced Motor Control. Participants who reported difficulty stopping nail-biting during strong urges (6/9) described that interventions that physically made it hard to move for nail-biting were acceptable and felt necessary. While some participants (4/6) explicitly requested stronger or punishment-like interventions, those with better self-control (3/9) preferred milder approaches. Despite these differences, participants consistently envisioned motor interference as a core component of future devices, with perceptual feedback (e.g., awareness reminders) serving as a baseline.

Autonomy in Device Control. Across participants, desired functions emphasized user control in intervention systems. Most (8/9) requested configurability and personalization, including adjustable intensity to their needs, mode selection like disabling notification in public, and linking multiple fingers to discourage "*switch biting between nails*" (P4). All participants also advocated for features that combine reminders with motor interference when urges escalate, as P1 noted "*different people bite for different reasons, so it needs to be more versatile.*" Furthermore, participants emphasized defaults favoring lower initial intensity and gradual scaling, not only for safety but to preserve a sense of control and sustained adherence, as P6 described, "*If it's too strong, people will take it off.*"

Usage Contexts and Social Concerns. Most participants (8/9) would use a future device in daily life, particularly in private settings (6/9), such as when alone at home, as P4 cited, "*If my roommates*

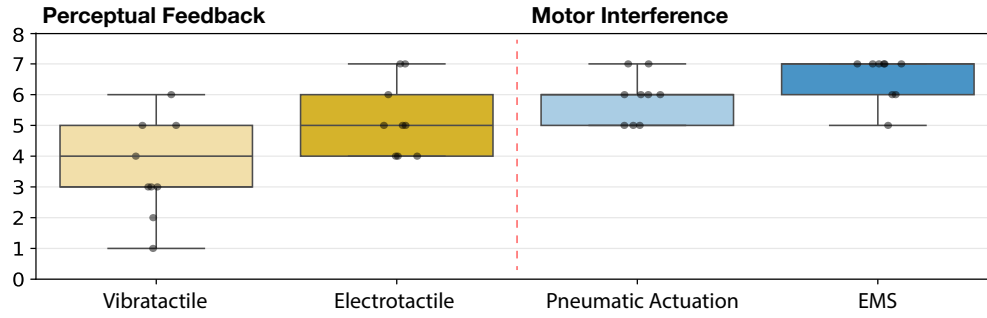


Figure 2: Self-reported perceived impact on nail-biting urges during simulated interactions with each probe (7-point Likert; 1=strongly disagree, 7=strongly agree). Boxes indicate medians and the interquartile range (IQR); whiskers extend to $1.5 \times$ IQR.

are around, I don't bite my nails, so I'll use it when I'm alone." Participants anticipated use during common triggers (e.g., anxiety, stress, boredom) and cognitively demanding tasks, especially when hands were idle. Participants also raised concerns about social conspicuousness when using the device in social situations, particularly with "pump noise" (P7) and potential "muscle fatigue" (P2) with EMS.

Form Factor and Wearability Requirements. Participants prioritized concealability and social acceptability, favoring low-profile forms that could be hidden under clothing or read as everyday accessories, as P7 noted, "skin-toned to blend with the body." They also emphasized wearability that does not cover fingertips or interfere with daily hand use, as well as ease of donning and doffing, with P9 noting, "fast attachment and removal." Beyond discreteness and convenience, participants sought designs that balance interruption with comfort. Soft, compliant constraints were favored for gently increasing effort without causing sharp sensations.

5 Discussion and Future Work

Rethinking Agency in Nail-biting Interventions. Our findings shed light on how user agency is experienced in nail-biting interventions. Participants with limited self-control described motor interference not as an undesirable loss of sense of agency, but as a pragmatic means to interrupt the hand-to-mouth action, contrasting it with perceptual feedback that primarily relies on voluntary inhibition. Notably, this acceptance was conditional. Participants distinguished between a sense of agency in momentary movement and autonomy over the intervention system, emphasizing the latter for acceptability. This distinction points to nail-biting interventions that allow users to authorize and configure temporary motor interference, rather than enforcing continuous or uncontrollable intervention. By preserving user control, such designs may better align with users' lived experiences of nail-biting urges. Future work should examine whether this conditional acceptance persists beyond brief exposure and what forms of configurability best support everyday use in nail-biting contexts.

Beyond Comfort as a Primary Design Objective. Wearable intervention research often treats comfort as a prerequisite for long-term adherence [26]. In our exploration, participants weighed comfort against perceived interruptive impact, particularly during high-urge moments. This suggests comfort not as a prerequisite, but

as a negotiable parameter that users may trade for stronger momentary interruption when needed. Future interventions might balance salience with everyday acceptability rather than maximizing comfort alone, for example by maintaining less intrusive feedback as a baseline while making stronger motor interference available during high-urge moments.

Ethical Implications of Interventions Perceived as Punitive. Participants' openness to punitive interventions raises important ethical considerations. Beyond momentary aversiveness, concerns include the potential normalization of self-directed harm and the influence of stigma on what users consider necessary [31]. Moreover, because nail-biting can be triggered by negative affect (e.g., stress or anxiety), punitive feedback may risk amplifying distress during emotionally vulnerable moments. These concerns highlight the importance of safeguards that govern how autonomy is exercised, such as constraints on intensity, deliberate friction for escalation, and the ability to pause interventions during heightened distress. Future work should assess how users employ such controls over time, how interventions are experienced in relation to stigma, and whether certain contexts warrant additional support.

Limitations. We note that our findings reflect participants' experiences during simulated nail-biting, as the study prioritized controlled exploration of user experience across multiple probes rather than measuring behavioral change or effectiveness in naturalistic contexts. Moreover, while the probe order was randomized, some influence of order or habituation may remain. Future work should evaluate integrated wearable systems in longitudinal deployments to assess sustained wearability, social acceptability, adherence, habituation, and behavioral outcomes across diverse nail-biting contexts.

6 Conclusion

We characterized how perceptual feedback and motor interference shape user experiences in nail-biting interventions. Through exploration sessions with active nail-biters, we found that participants often preferred motor interference despite a reduced sense of agency, acknowledging their limited self-control. However, they emphasized the importance of retaining autonomy over how the intervention system operates. This distinction between a momentary sense of agency and desired autonomy over the system offers

design insights for acceptable wearable nail-biting interventions and motivates future longitudinal exploration in everyday contexts.

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